

Nucleic Acids: Structure, Function & Biotechnological Applications

B.Tech Biotechnology – Nucleic Acid Fundamentals

Introduction to Nucleic Acids

Nucleic acids are biological macromolecules found in all living cells, serving as the blueprint of life. First isolated by Friedrich Miescher in 1869 from nuclei of pus cells, these molecules were initially called "nuclein." Today, we recognize two major types: DNA and RNA, which store, transmit, and express genetic information.

The discovery of nucleic acids revolutionized our understanding of heredity and molecular biology. Every living organism—from bacteria to humans—depends on these molecules to encode and propagate genetic information across generations.

Key Discoveries

- Miescher (1869): First isolation from cell nuclei
- Chargaff (1950): Base pairing rules
- Watson & Crick (1953): Double helix structure
- Franklin: Critical X-ray diffraction data

Structure of Nucleotides

Nucleotides are the building blocks of nucleic acids, consisting of three components linked together. These components form the foundation of genetic information storage and transmission.



Pentose Sugar

Deoxyribose in DNA or ribose in RNA forms the structural backbone



Phosphate Group

PO_4^{3-} provides negative charge and links sugars together



Nitrogenous Base

Carries genetic information through specific base pairing

Nucleotides connect via **phosphodiester bonds** to form long chains called polynucleotides. The sugar-phosphate backbone creates the structural framework, while nitrogenous bases project inward to encode information.

Nitrogenous Bases: Purines & Pyrimidines

Purines (Double-Ring)

These bases have a double-ring structure and include:

- **Adenine (A)** - pairs with thymine/uracil
- **Guanine (G)** - pairs with cytosine

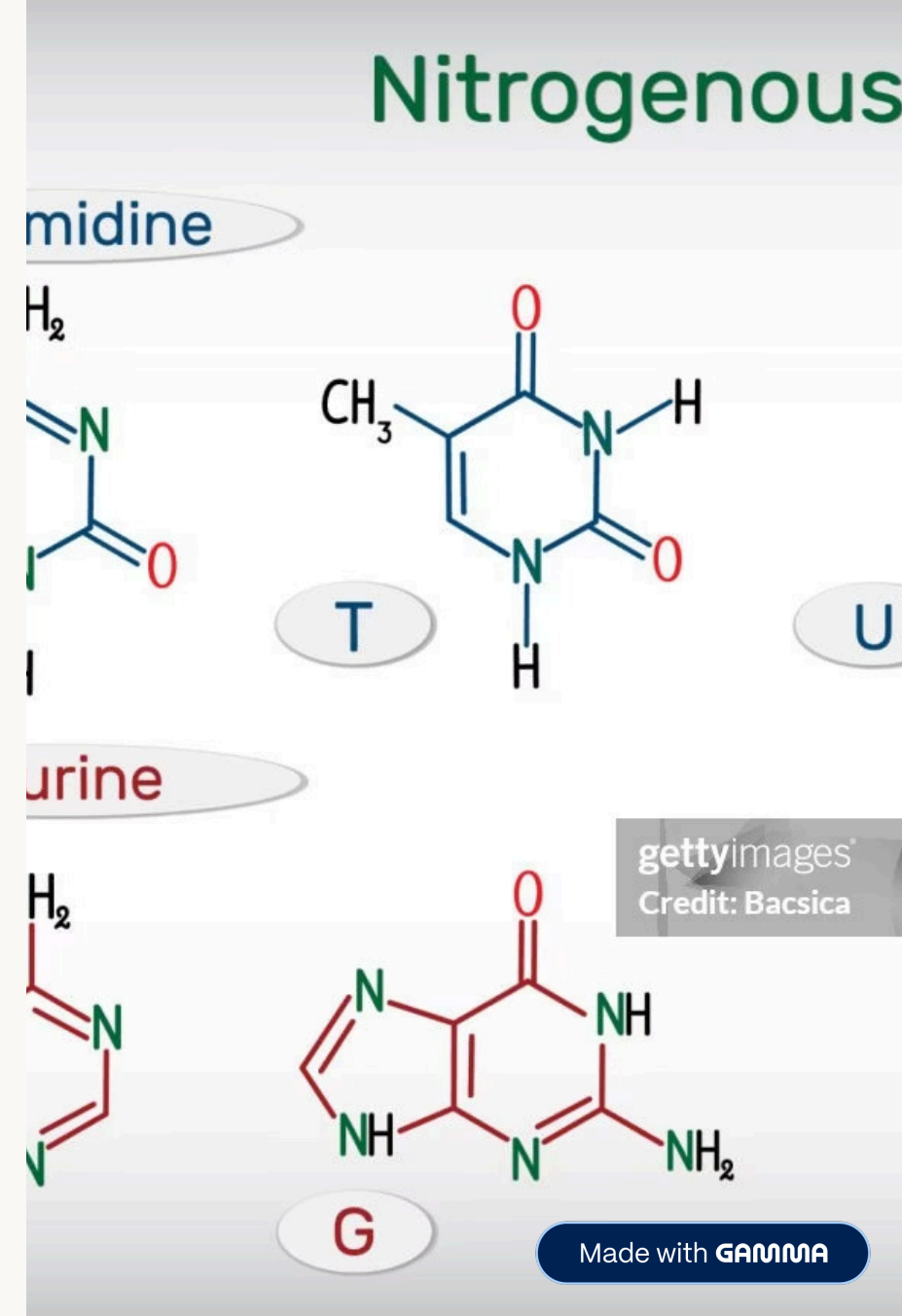
Pyrimidines (Single-Ring)

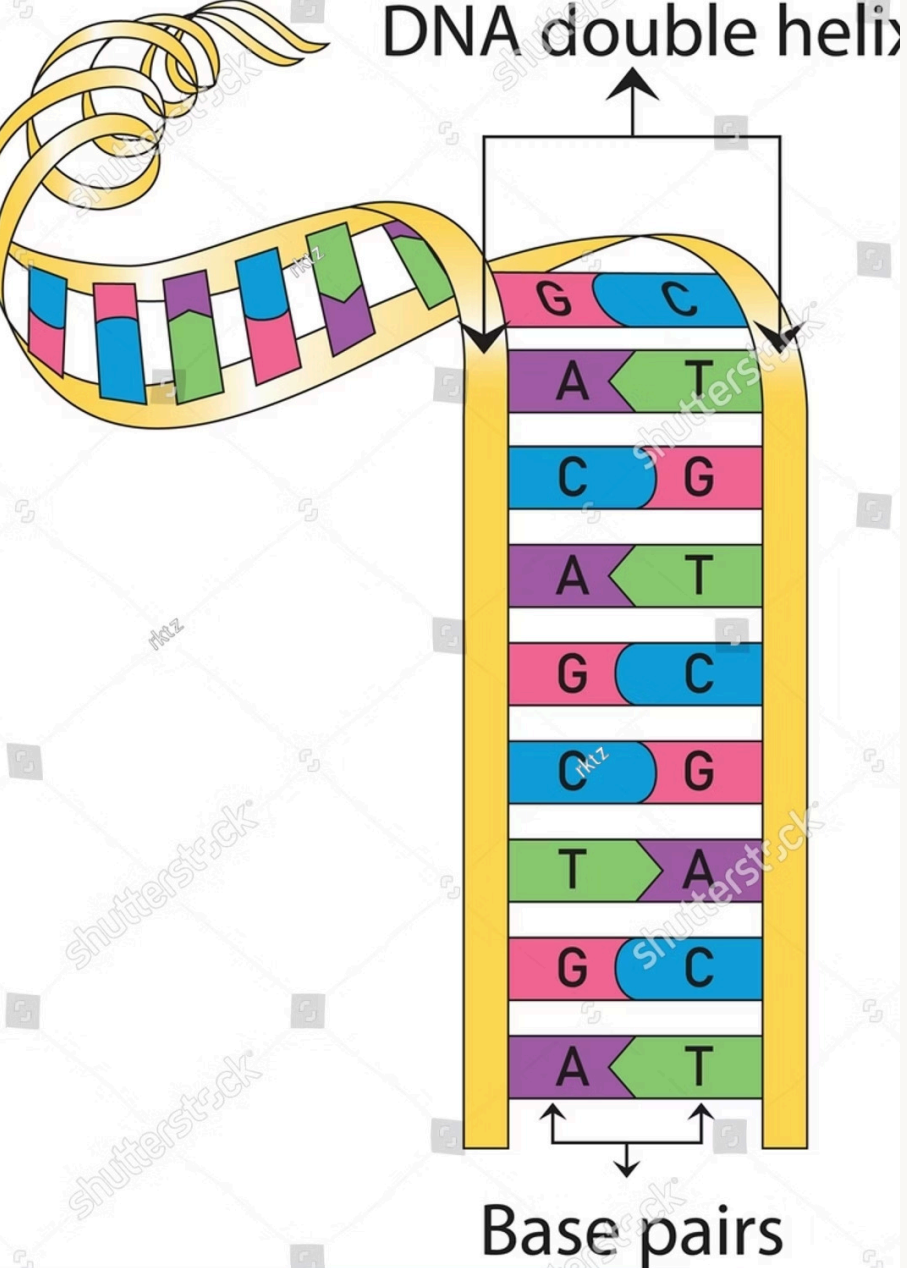
These bases have a single-ring structure and include:

- **Cytosine (C)** - pairs with guanine
- **Thymine (T)** - DNA only, pairs with adenine
- **Uracil (U)** - RNA only, replaces thymine

Chargaff's Rule

In DNA, the amount of adenine equals thymine ($A = T$), and guanine equals cytosine ($G \equiv C$). This complementary base pairing was crucial for understanding DNA replication.





DNA Structure: The Double Helix

01

Right-Handed Double Helix

Proposed by Watson & Crick in 1953, based on Rosalind Franklin's X-ray diffraction data

02

Antiparallel Strands

Two strands run in opposite directions ($5' \rightarrow 3'$ and $3' \rightarrow 5'$)

03

Complementary Base Pairing

A — T forms 2 hydrogen bonds, G ≡ ≡ ≡ C forms 3 hydrogen bonds

04

Dimensions

Diameter: ~2 nm, one full turn: ~3.4 nm (10 base pairs)

RNA: The Versatile Molecule

Unlike DNA, RNA is typically single-stranded and uses ribose sugar with uracil replacing thymine. RNA plays diverse roles in gene expression and protein synthesis.



Messenger RNA (mRNA)

Carries genetic code from DNA to ribosomes for protein synthesis.
Contains codons that specify amino acid sequences.



Transfer RNA (tRNA)

Brings amino acids to ribosomes during translation. Contains anticodon that pairs with mRNA codons.



Ribosomal RNA (rRNA)

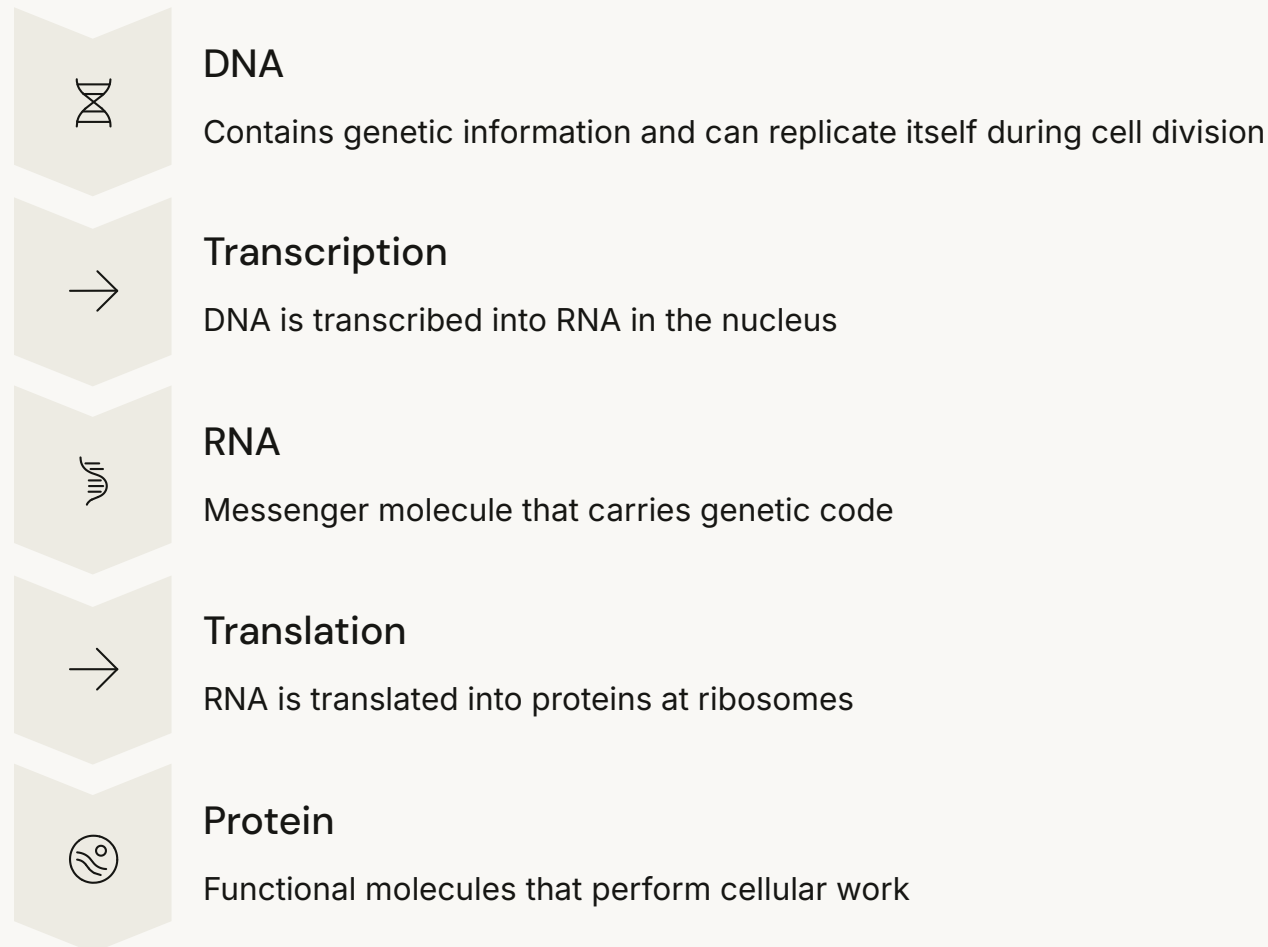
Forms structural and catalytic core of ribosomes. rRNA acts as a ribozyme with enzymatic activity.

DNA vs RNA: Key Differences

Feature	DNA	RNA
Full Name	Deoxyribonucleic Acid	Ribonucleic Acid
Structure	Double-stranded helix	Single-stranded (mostly)
Sugar	Deoxyribose (C ₂ -H)	Ribose (C ₂ -OH)
Bases	A, G, C, T	A, G, C, U
Base Pairing	A=T, G≡C	A=U, G≡C
Stability	Highly stable	Less stable (degrades faster)
Location	Nucleus (primarily)	Nucleus + Cytoplasm
Primary Function	Long-term genetic storage	Protein synthesis, regulation
Replication	Self-replicating	Synthesized from DNA template
Length	Very long (millions of bp)	Relatively short

Central Dogma of Molecular Biology

Proposed by Francis Crick in 1958, the Central Dogma describes how genetic information flows in biological systems. This framework defines the fundamental processes of molecular biology.



Reverse Transcription

Discovered in retroviruses like HIV, reverse transcriptase enables RNA → DNA information flow. This exception to the original dogma powers RT-PCR and cDNA library construction.

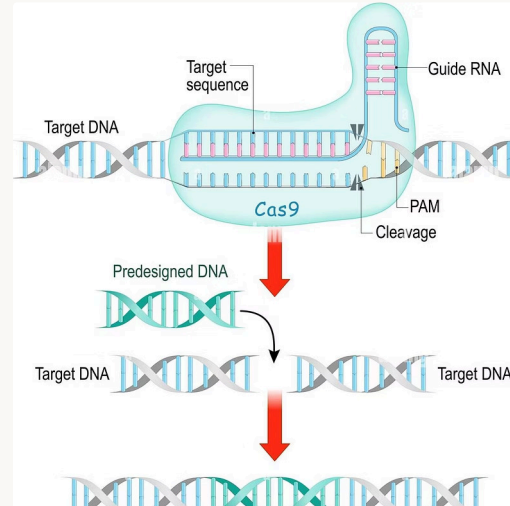
Biotechnological Applications

Understanding nucleic acids has enabled revolutionary technologies that transform medicine, agriculture, and research.



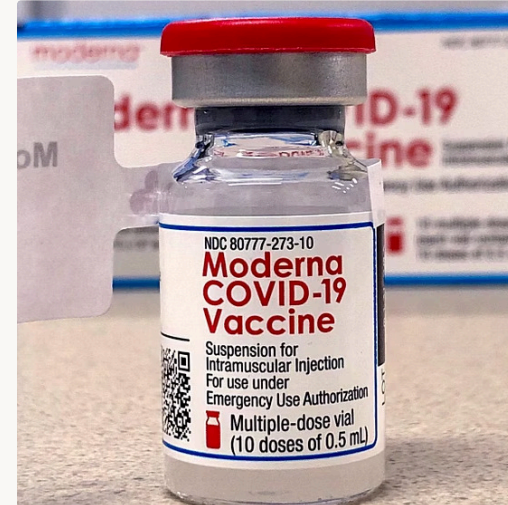
PCR Amplification

Polymerase Chain Reaction amplifies specific DNA segments. Applications include diagnostics, forensics, and COVID-19 testing.



CRISPR-Cas9

Precise gene editing using guide RNA and Cas9 protein. Applications include gene therapy, crop improvement, and genetic disease treatment.



mRNA Vaccines

Synthetic mRNA delivers instructions to cells, training immune systems. Pfizer and Moderna COVID-19 vaccines saved millions of lives.



GM Crops

Genetic engineering creates crops with improved traits. Examples include Bt cotton (pest resistance) and Golden Rice (vitamin A enrichment).

Conclusion: Key Takeaways

1 Nucleic acids are fundamental

DNA and RNA serve as molecules of heredity in all living organisms

2 Structure determines function

DNA's double helix provides stability for long-term storage;
RNA's single strand enables versatility

3 The Central Dogma guides information flow

Understanding DNA → RNA → Protein transformations is essential for biotechnology

4 Applications are transformative

PCR, CRISPR, mRNA vaccines, and GM crops demonstrate real-world impact

"The secret of life is a chemical called deoxyribonucleic acid." — *Francis Crick*

Mastering nucleic acid fundamentals provides the foundation for all biotechnology work, from research to clinical applications.